MALAYALAM MOVIE REVIEW MULTI LABEL ASPECT BASED CLASSIFICATION

a machine learning project that explores multi label classification of Malayalam movie review using TF-|IDF extraction and traditional machine learning models

OBJECTIVE

To investigate whether machine learning models can predict nine aspect annotations from Malayalam movie review

Aspects are:

1) Direction
2) Acting
3) Songs
4) Screenplay
5) Story
6) Camera
7) Casting
8) BGM
9) Editing

This project compares word level and character level TF-IDF using Logistic Regression and Linear Support Vector Machines(SVM)

Dataset

The project uses **MABSA** Malayalam movie review dataset the dataset contains 4000 reviews and nine aspects the dataset is not included in the dataset users must have to ensure the permission to use it

METHODOLOGY

1) Data inspection and preprocessing
2) Removal of empty and Duplicate reviews
3) Preparation of multi label targets
4) Exploratory data analysis
5) Iterative stratification for train-test splitting
6) TF-IDF feature extraction
7) Model training and evaluation
8) Error analysis and feature inspection

Models and Experiments

1) Default word-level TF-IDF + Logistical Regression
2) Default word-level TF-IDF + Linear SVM
3) Malayalam-aware word-level TF-IDF + Logistic Regression

4) Malayalam-aware word-level TF-IDF + Linear SVM
5) Character-level TF-IDF + Linear SVM

RESULTS

MODEL	EXACT MATCH ACCURACY	HAMMING LOSS	MICRO F1	MACRO F1
Default word TF-IDF + Logistic Regression	**40.06%**	**11.06%**	**76.00%**	**76.00%**
Default Word TF-IDF + Linear SVM	**49.19%**	**9.44%**	**79.58%**	**79.29%**
Malayalam Word TF-IDF + Logistic Regression	**54.49%**	**7.58%**	**83.34%**	**82.96%**
Malayalam Word TF-IDF + Linear SVM	**63.48%**	**6.37%**	**86.12%**	**85.90%**
Character TF-IDF + Linear SVM	**63.62%**	**5.96%**	**87.40%**	**87.02%**

The **character level TF-IDF with Linear SVM** achieved the strongest overall results in this experiment.

These results are from a single 80-20 iterative stratifies train-test split using random state=42

Label Limitations

The original dataset contains values **1.0,0.0,0.5 and missing values**

For this experiment.

- Non missing values are mapped to 1
- Missing values are mapped to 0

The tasks predicts whether an aspect has a recorded annotations.it **does not** predict positive, negative or neutral sentiment. Missing annotations may not means that the aspect does not exists in the review this is an important limitations.

Technologies Used

- Python
- Pandas
- NumPy
- Scikit-learn
- TF-IDF
- Logistic Regression
- Linear SVM
- Iterative Stratification
- Matplotlib
- Seaborn
- Google Colab

Limitations and Future Work

Current limitations:

- Missing annotations are treated as negative labels.
- Sentiment polarity is not modelled
- Reviews with zero recorded aspect are excluded from modelling
- Evaluation uses only one train test split
- The Malayalam tokenizer is a simple Unicode-based tokenizer rather than a full linguistic or morphological tokenizer

Possible Future improvements:

- Investigate missing-label handling.
- Explore combined word level and character level feature.
- Evaluate models using repeated splits or cross validation.
- Explore Malayalam-specific pretrained language models.
- Investigate aspect level sentiment classification.

Conclusion

This project investigates traditional machine-learning approaches for multi-label aspect annotation prediction in Malayalam movie reviews. The experiments highlight the importance of text representation, particularly character-level TF-IDF, for this dataset.

The findings are preliminary and require further validation.